YEAR 9 PHYSICS, AQA GCSE, UNIT 1, PART 1

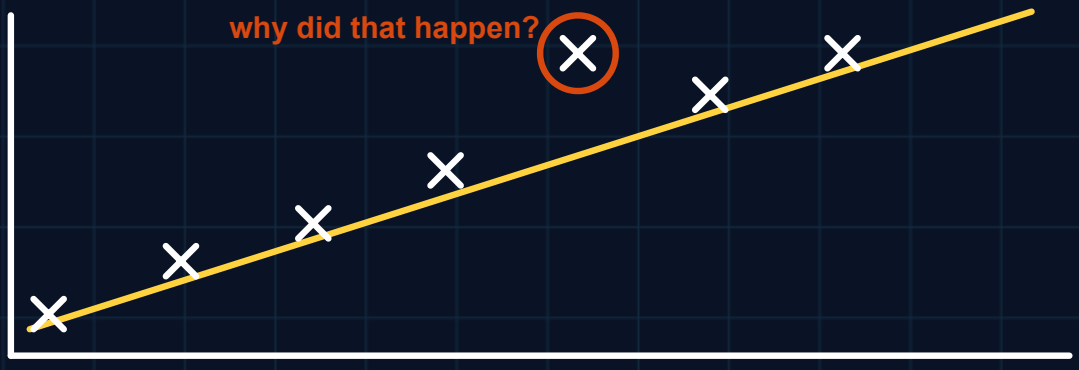
WORK LIKE A PHYSICIST

PART 1 OF 3

LESSONS 1–2: CAN THIS DATA BE TRUSTED?; WHAT ARE YOU ACTUALLY CHANGING?

2 LESSONS **1** PRACTICAL **20** KEY WORDS

> *"The first principle is that you must not fool yourself, and you are the easiest person to fool."*
>
> RICHARD FEYNMAN, PHYSICIST



NAME _____ CLASS _____

TEACHER _____ LAB PARTNER _____

START HERE

HOW THIS BOOKLET WORKS

Anyone can have an opinion about how the world works. A physicist has to do something harder: produce **evidence** that convinces people who were not there. This unit teaches you how. Keep the booklet: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
The explanation, diagrams and worked examples. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own.

PRACTICAL
Plan, record your measurements and work out results from them.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: The thing you change is the _____ **(independent / dependent / control)** variable. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The thing you measure is the

independent	dependent	control

variable.

THE ROUTE THROUGH THIS PART

#	LESSON	PRACTICAL
01	**Can this data be trusted?** Precision and accuracy, random and systematic error, uncertainty, resolution	Measure one thing three times
02	**What are you actually changing?** Independent, dependent and control variables; continuous and categoric data; choosing a graph	Class demo: which surface slows the car most?

This booklet is Part 1 of 3: Lessons 1–2. The **Physicist's toolkit** on the last page has every rule, calculation and checklist you need.

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Lesson

01 CAN THIS DATA BE TRUSTED?

AQA Working scientifically WS 2.6, 3.4, 3.7; Physics and Combined Science

BIG QUESTION **Thousands of repeat measurements agreed perfectly, and the result was still wrong. How can that happen?**

BY THE END YOU CAN

- tell the difference between precise and accurate
- describe random and systematic errors, and say why repeats only help with one of them
- calculate a mean and an uncertainty from repeat readings
- say what the resolution of an instrument is

precise **accurate** **random error** **systematic error** **uncertainty** **resolution**

THE CASE (2011)

The loose cable that broke physics. Physicists in Italy timed neutrinos travelling 730 km from Switzerland. They seemed to arrive about 60 nanoseconds before light could. Thousands of repeats agreed. Months later the team found a fibre-optic cable that was not properly connected. It shifted every reading by the same amount, so repeating could never reveal it, and the result was withdrawn.

Consistent does not mean correct.



REMEMBER

WHAT DO YOU ALREADY THINK?

No wrong answers: this is your starting point.

1

You time a toy car rolling down a ramp three times and get three different times. Give one reason why.

2

Calculate the mean of 4 cm, 5 cm and 6 cm.

3

Which ruler can measure more finely?

(the one marked in centimetres / the one marked in millimetres)

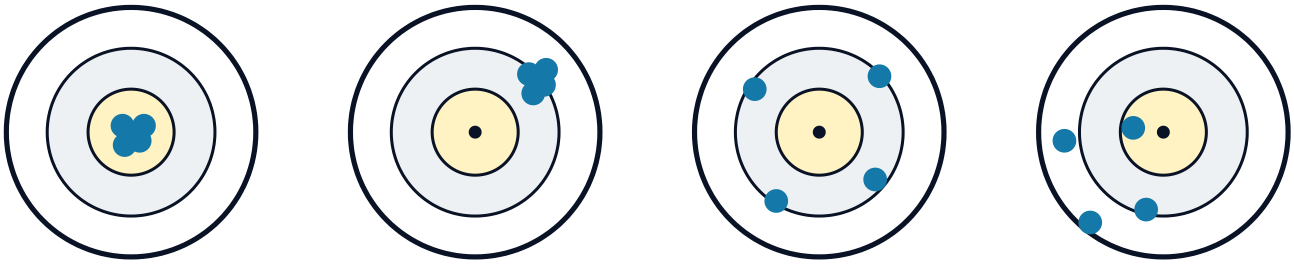
EXPLAIN

CAN YOU TRUST A MEASUREMENT?

Every measurement has some doubt in it. Measure the same thing three times and you rarely get the same number three times. A physicist asks: **how far can I trust this result?**

PRECISE AND ACCURATE ARE DIFFERENT WORDS

WORD	MEANING	HOW YOU CAN TELL
Precise	repeat readings are **close to each other**	look at the spread of the readings
Accurate	a reading is **close to the true value**	compare with the true value, e.g. weigh a known 100 g mass

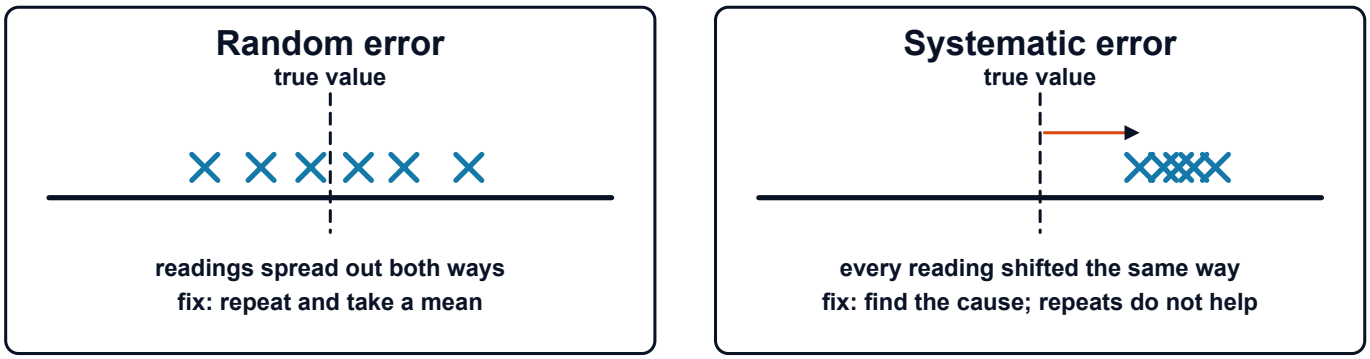


Precise and accurate | **Precise, but not accurate** | **Accurate on average, not precise** | **Neither precise nor accurate**

Centre = the true value. Blue dots = repeat readings.

! **You can be perfectly precise and completely wrong.** A tight group of readings looks trustworthy, so nobody checks it. That is why precise-but-wrong is the dangerous one.

TWO KINDS OF ERROR

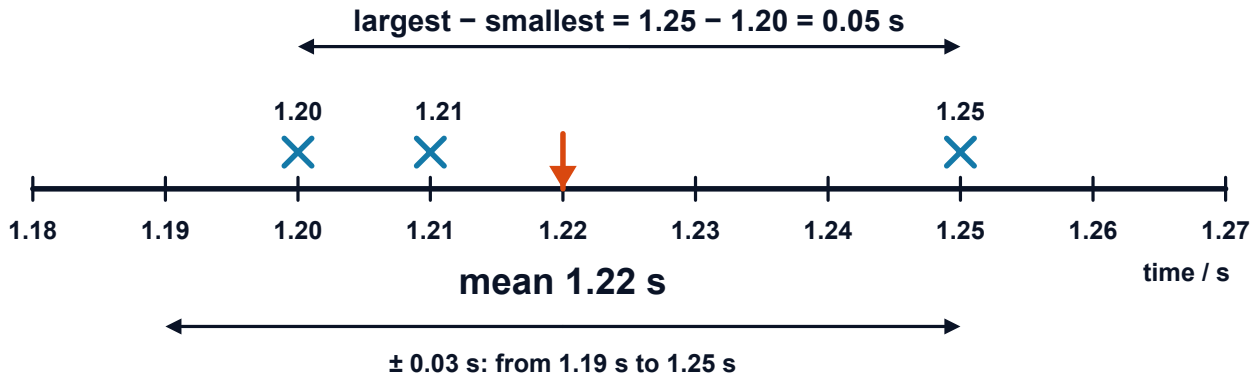


ERROR	WHAT IT DOES	EXAMPLE	WHAT TO DO
Random	spreads readings out unpredictably: some too high, some too low	reaction time with a stopwatch; judging a bounce height by eye	repeat and take a mean; use better equipment, such as light gates
Systematic	shifts every reading by the same amount, in the same direction	a balance that reads 0.2 g with nothing on it (a **zero error**); measuring from the end of a ruler, not the zero mark	find the cause: check the zero, check against a known value. Repeats do **not** help

HOW TO TELL WHICH KIND YOU HAVE

Ask: would this make the reading wrong by the same amount, in the same direction, every time? **Yes:** systematic. **Up one time and down the next:** random. A known systematic error can be corrected: if the balance reads 0.2 g with nothing on it, subtract 0.2 g from every reading.

UNCERTAINTY: PUT A NUMBER ON THE DOUBT



WORKED EXAMPLE

Three readings: 1.20 s, 1.25 s, 1.21 s. Find the mean and the uncertainty.

MEAN	(1.20 + 1.25 + 1.21) ÷ 3 = 3.66 ÷ 3 = 1.22 s
LARGEST − SMALLEST	1.25 − 1.20 = 0.05 s
UNCERTAINTY	0.05 ÷ 2 = 0.025 s, about 0.03 s
RESULT	**1.22 s ± 0.03 s**

YOUR TURN

Three readings: 4.6 cm, 5.0 cm, 4.8 cm. Find the mean and the uncertainty.

MEAN	
LARGEST − SMALLEST	
UNCERTAINTY	
RESULT	

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! **uncertainty = (largest − smallest) ÷ 2.** Why divide by 2? Largest − smallest is the whole width of the spread. The mean sits near the middle, so the readings reach about half that width above it and half below. Read “1.22 s ± 0.03 s” as “about 1.22 s, give or take 0.03 s”. The **range** of these readings is from 1.20 s to 1.25 s: a range is stated “from … to …”, never as one number.

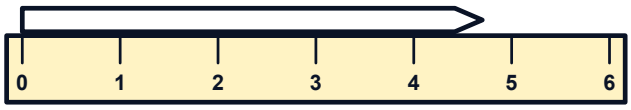
RESOLUTION: WHAT CAN THE INSTRUMENT SEE?

The **resolution** of an instrument is the smallest change it can detect. A ruler marked in millimetres has a resolution of 1 mm. A stopwatch that reads to 0.01 s has a resolution of 0.01 s.

Match the instrument to the job: 1 mm is tiny next to the length of a desk, but it is half the thickness of a coin.

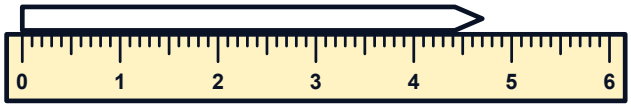
! **Trap.** Better resolution gives you more decimal places. It does *not* remove a systematic error: a balance with a zero error gives a precise wrong answer to two decimal places.

A: marked every 1 cm



Reading: about 5 cm resolution 1 cm

B: marked every 1 mm



Reading: 4.7 cm resolution 1 mm

Repeatable: you do it again with the same method and equipment and get similar results. **Reproducible:** another person, or different equipment, gets similar results. Reproducible is the stronger test.

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PRACTISE

MEASUREMENT DETECTIVE

1 Three students timed the same toy car rolling down the same ramp.

Set	Trial 1 / s	Trial 2 / s	Trial 3 / s	Mean / s
A	1.20	1.23	1.19	1.21
B	1.05	1.47	1.12	1.21
C	0.90	0.93	0.89	0.91

a Sets A and B have the same mean. The more precise set is **A** / **B** Explain how you can tell.

b The true time is about 1.21 s. Set C is precise but **accurate** / **not accurate** Suggest what could shift every reading in set C by the same amount.

c What is the point of taking three readings instead of one?

2 A balance is used to weigh a mass of exactly 50.0 g three times. Draw **one** line from each set of readings to the description that fits.

50.1 g, 49.9 g, 50.0 g	precise but not accurate
52.4 g, 52.5 g, 52.4 g	neither precise nor accurate
47.0 g, 53.1 g, 49.8 g	precise and accurate
44.2 g, 46.9 g, 41.7 g	accurate on average, not precise

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3 The neutrino readings in *The case* were

precise and accurate	precise but not accurate	accurate but not precise

Explain your choice.

4 SUPPORTED Complete the table. All the readings are in cm. Then answer the questions below it.

Set	Trial 1	Trial 2	Trial 3	Mean	Largest − smallest	Uncertainty
A	12.0	12.4	12.1			
B	8.4	9.1	8.6			
C	24.9	25.1	25.1			
D	3.20	3.65	3.25			

a Show your working for the mean of one set.

b The most precise set is

A	B	C	D

The set with the largest uncertainty is

A	B	C	D

c One reading in set D may be a mistake. Which one? What would you do before using the mean?

5 Is each problem a random or a systematic error? Draw a ring around one word in each row.

Reaction time: the stopwatch is started sometimes early, sometimes late.

random	systematic

The ruler is read from its very end, not from the zero mark.

random	systematic

A balance reads 0.2 g with nothing on it.

random	systematic

The height of a fast bounce is judged by eye.

random	systematic

A draught from an open window comes and goes.

random	systematic

A scale is always read from above, at the same angle.

random	systematic

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6 Explain why taking more repeats cannot remove a systematic error.

7 Complete the table.

Instrument	Resolution	Could it show a change of 0.5 mm?
Ruler marked every 1 cm	1 cm / 1 mm / 0.1 mm	yes / no
Ruler marked every 1 mm	1 cm / 1 mm / 0.1 mm	yes / no
Digital calipers reading to 0.1 mm	1 cm / 1 mm / 0.1 mm	yes / no

A stopwatch reads to 0.01 s. Its resolution is | **0.01 s** | **0.1 s** | **1 s** |

8 Explain what is wrong with what Sam says.

Sam My digital balance shows two decimal places, so my results must be accurate.

9 CHALLENGE Data detective. For each claim, write what you would need to know before believing it, and the error you suspect.

The claim	What I would need to know first	Error I suspect
"Our reaction times improved after training." The two tests were timed by different people.		random / systematic
"The room is exactly 6.000 m long." Measured once, with the tape held slightly slack.		random / systematic
"This spring stretches 4.0 cm for each 100 g." One reading was taken for each mass.		random / systematic

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10 Checkpoint. Complete the sentences.

Choose answers from the box. Each word may be used once, more than once or not at all.

precise **accurate** **random** **systematic** **resolution**

A set of results that are close together is _____.

A result that is close to the true value is _____.

Repeats reduce the effect of _____ errors, but not _____ errors.

The smallest change an instrument can detect is its _____.

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INVESTIGATE

MEASURE ONE THING THREE TIMES

Your teacher gives you one task: a dropped-ruler reaction test, a toy car on a ramp, the bounce of a ball, or one object measured with different rulers.

1. Agree exactly what you will measure, and from where to where.
2. Take three readings. Write each one down as soon as you take it.
3. Calculate the mean and the uncertainty.

What I measured (with unit)	Trial 1	Trial 2	Trial 3	Mean	Uncertainty

Instrument I used	
Its resolution	

11. Name one random error that affected you. How do you know it was random?

12. Name one possible systematic error. How would you look for it?

13. One thing I would do differently to make my readings more repeatable:

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: Show the sum before you divide. Give the answer to the same number of decimal places as the readings.

14 A student times a pendulum swinging 10 times. The readings are 15.2 s, 15.6 s and 15.3 s. Calculate the mean time. **[2 marks]**

15 Calculate the uncertainty in the mean. Use uncertainty = (largest − smallest) ÷ 2 **[2 marks]**

16 The student always starts the stopwatch late, by about the same amount. What type of error is this? Tick **one** box. **[1 mark]**

☐ random error ☐ systematic error ☐ anomalous result

17 Explain why repeating the measurement would not reduce this error. **[1 mark]**

TAKE IT HOME

18 Find a claim with a number in it: an advert, a headline or a sports statistic. Write it down, then write one question you would have to ask before you believed it. Bring it to Lesson 2.

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REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- **Precise:** repeat readings close together. **Accurate:** close to the true value.
- **Random error** spreads readings both ways; repeats and a mean reduce it.
- **Systematic error** shifts every reading the same way; repeats do not help.
- uncertainty = (largest − smallest) ÷ 2. Write the result as mean ± uncertainty.
- **Resolution:** the smallest change an instrument can detect.

COVER THE PAGE AND ANSWER

1 What does precise mean?

2 What does accurate mean?

3 How do you calculate the uncertainty from repeats?

4 Which kind of error do repeats reduce?

5 A balance reads 0.2 g with nothing on it. What kind of error?

6 What is the resolution of a ruler with mm marks?

I CAN…

- tell precision from accuracy
- calculate a mean and an uncertainty from repeats
- tell random from systematic error and say how to deal with each
- state the resolution of an instrument

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Lesson

02 WHAT ARE YOU ACTUALLY CHANGING?

AQA Working scientifically WS 2.2, 3.1; Physics and Combined Science

BIG QUESTION **An advert says a drink makes you smarter. What would you need to know before you believed it?**

BY THE END YOU CAN

- identify the independent, dependent and control variables in any investigation
- decide whether data is continuous or categoric
- choose a line graph or a bar chart, and give the reason

independent variable **dependent variable** **control variable** **continuous** **categoric** **fair test**

THE CASE

The study that proved nothing. An invented example. A drinks company says students who drank its product scored higher in a maths test. But the drinkers sat the test in the morning and the others after lunch, in a different room. Two things changed at once, so the result cannot tell you which one made the difference.

Change one thing. Measure one thing. Keep everything else the same.

REMEMBER

DO NOW

From earlier lessons. No notes!

1
Precise means repeat readings are _____ **(close to each other / close to the true value)**.

2
Repeats and a mean reduce the effect of _____ **(random / systematic)** errors.

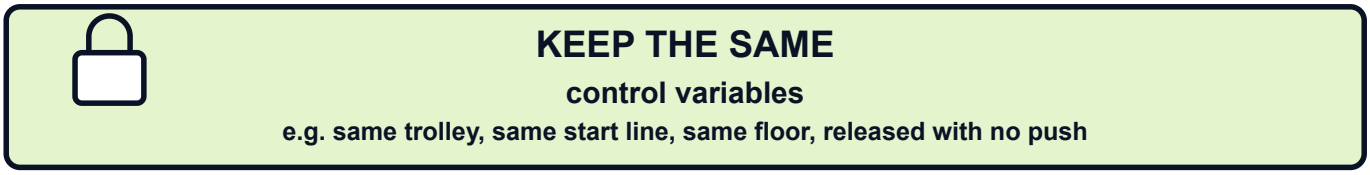
3
Calculate the uncertainty for 3.2 s, 3.6 s and 3.4 s.

EXPLAIN

TAKE THE QUESTION APART

Every investigation answers one question: **How does X affect Y?** X is what you change. Y is what you measure.

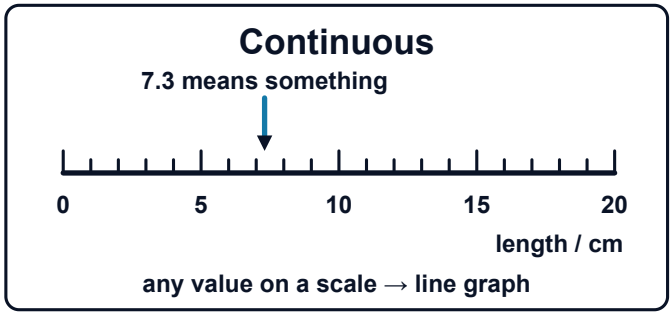


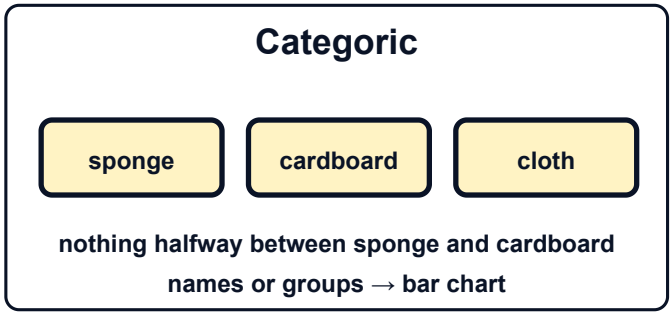


VARIABLE	MEANING	EXAMPLE: THE RAMP
Independent	the **one** thing you change on purpose	ramp height
Dependent	the thing you **measure**	distance the trolley travels
Control	everything you **keep the same**, so it cannot affect the result	same trolley, same start line, same floor, no push

! **Change only one thing at a time.** Test a longer pendulum with a heavier bob, and the swing takes longer: was it the length or the mass? You cannot say. “Keep it a fair test” earns nothing on its own: name the variable and how you keep it the same, such as “use the same bob each time”.

TWO KINDS OF DATA



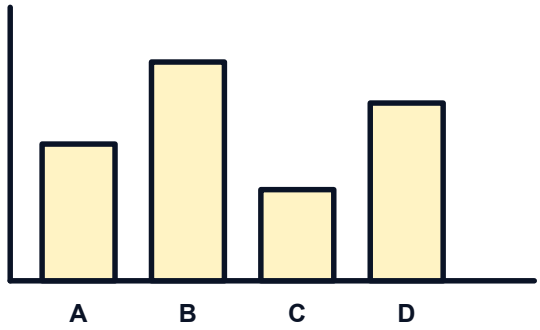


Test: is there a value in between? Between 10 cm and 20 cm there is 15 cm, and it means something: length is **continuous**. There is nothing halfway between sponge and cardboard: material is **categoric**. The same thing can be recorded either way: “hot, warm, cold” is categoric, but “60 °C, 40 °C, 20 °C” is continuous and tells you much more. When you can, measure.

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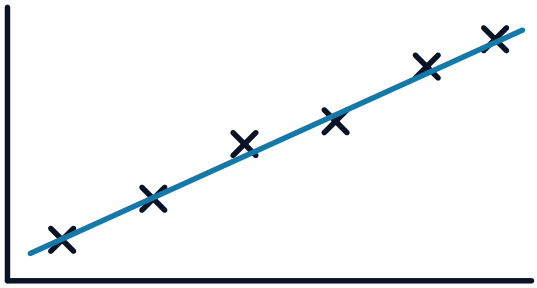
WHICH GRAPH?

Categoric → bar chart



separate bars with gaps

Continuous → line graph



crosses and a best-fit line

THE RULE THAT DECIDES YOUR GRAPH

Continuous independent variable → line graph. Categoric independent variable → bar chart.

A line says something about the values between your points, which only makes sense if in-between values exist. Separate bars show separate things.

! **The mistake almost everyone makes:** “My results are numbers, so it’s a line graph.” In “Which surface gives the most friction?” you measure a distance in centimetres, but surface type is what you changed, and surfaces are categories. So it is a bar chart. Look only at the independent variable.

PRACTISE

DECODE ANY INVESTIGATION

1 Fill in all three columns. Take care with the controls: they are the column people rush.

Investigation question	I change (independent)	I measure (dependent)	I keep the same (two controls)
How does ramp height affect the speed of a trolley?			
Which material is the best thermal insulator?			
How does the number of elastic bands affect launch distance?			
Which drink contains the most sugar?			

Pick one of the four. If you did not keep your controls the same, what would go wrong with your conclusion?

2 Is each variable continuous or categoric? Draw a ring around one word in each row.

The mass of a ball **continuous** **categoric**

The colour of a car **continuous** **categoric**

The time a ball takes to fall **continuous** **categoric**

The type of metal in a wire **continuous** **categoric**

The temperature of the water **continuous** **categoric**

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3 Sort the investigations. Both columns are needed for full credit.

Investigation	Graph	The independent variable is…
How does temperature affect dissolving time?	line / bar	continuous / categoric
Which surface gives the most friction?	line / bar	continuous / categoric
How does mass affect the stretch of a spring?	line / bar	continuous / categoric
Which paper towel absorbs the most water?	line / bar	continuous / categoric
How does pendulum length affect the time of a swing?	line / bar	continuous / categoric
Which material is the best sound absorber?	line / bar	continuous / categoric

4 CHALLENGE Fix the broken plan. A student writes: *"I will test whether expensive tennis balls bounce better. I'll drop an expensive ball from the window and a cheap one from my hand, and see which bounces higher."*

a Name two things wrong with this plan.

b Rewrite it as a question in the form "How does X affect Y?"

c Your question needs a **bar chart** | **line graph** because the independent variable is **continuous** | **categoric**

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5 Checkpoint. Complete the sentences.

> Choose answers from the box. Each word may be used once, more than once or not at all.
>
> **change** **measure** **keep the same** **continuous** **categoric**

The independent variable is the thing I _____.

The dependent variable is the thing I _____.

Control variables are the things I _____.

Use a line graph when the independent variable is _____.

Use a bar chart when the independent variable is _____.

6 **EXPLAIN** Explain to someone who missed the lesson why the choice of graph depends on the independent variable and not on the dependent one.

INVESTIGATE

DEMO: WHICH SURFACE SLOWS THE CAR MOST?

Your teacher rolls the same car down the same ramp onto different surfaces: bench, carpet, rubber mat, sandpaper.

Independent variable	
Dependent variable (with unit)	
Graph type	**line graph** | **bar chart**

7 List three control variables. Put a star by the one that would be hardest to keep the same.

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AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: The independent variable is the one the student chooses to change.

8 A student investigates how the length of a wire affects its resistance. Complete the sentence. **[1 mark]**

The independent variable is

the length of the wire	**the resistance**	**the temperature**

9 Give **one** control variable for this investigation. **[1 mark]**

10 Complete the sentence. **[1 mark]**

The results should be shown on a | **bar chart** | **line graph** | because the length of the wire is | **categoric** | **continuous** |

11 Which investigation should be shown on a bar chart? Tick **one** box. **[1 mark]**

- ☐ how the height of a drop affects the time a ball takes to fall
- ☐ how the type of shoe affects grip
- ☐ how temperature affects how fast sugar dissolves
- ☐ how mass affects the stretch of a spring

TAKE IT HOME

12 Write five investigation questions of your own in the form “How does X affect Y?”: at least two that need a bar chart and two that need a line graph. For each, name the independent variable and say whether it is continuous or categoric.

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REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- **Independent:** the one thing you change. **Dependent:** what you measure. **Control:** what you keep the same.
- Change only one thing, or you cannot tell what caused the result.
- **Continuous:** any value on a scale. **Categoric:** names or groups.
- Continuous independent variable → line graph. Categoric → bar chart.

COVER THE PAGE AND ANSWER

1 What is the dependent variable?

2 Why must control variables be kept the same?

3 Is surface type continuous or categoric?

4 Which variable decides the type of graph?

I CAN…

identify the independent, dependent and control variables

say whether data is continuous or categoric

choose a line graph or a bar chart and give the reason

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REFERENCE

KEY WORDS

Learn the meaning and the spelling. In exams, precise words earn marks: say “random error”, never “human error”.

Accurate L1: Close to the true value

Precise L1: Repeat readings are close to each other: little spread about the mean

Trial L1: One repeat of a measurement

Mean L1: The total of the trials divided by the number of trials

Uncertainty L1: The interval within which the true value can be expected to lie. From repeats: (largest − smallest) ÷ 2

Range L1: The maximum and minimum values of the independent or dependent variables, stated “from … to …”

Resolution L1: The smallest change a measuring instrument can detect

Random error L1: Unpredictable spread between repeats. Repeats and a mean reduce it

Systematic error L1: An error that shifts every reading by the same amount in the same direction. Repeats do not fix it

Zero error L1: An instrument reads a value when it should read zero: a systematic error

Repeatable L1: The same person, method and equipment give similar results again

Reproducible L1: Another person, or different equipment or method, gives similar results

Independent variable L2: The one thing you change

Dependent variable L2: The thing you measure

Control variable L2: Something you keep the same so it cannot affect the result

Fair test L2: Only the independent variable changes; all the control variables are kept the same

Continuous data L2: Numbers that can take any value on a scale, such as length, time or mass

Categoric data L2: Names or groups, such as material, colour or brand

Line graph L2: The graph to use when the independent variable is continuous

Bar chart L2: The graph to use when the independent variable is categoric

PHRASES THAT PICK UP MARKS

INSTEAD OF WRITING…	WRITE…
“It was human error.”	“Reaction time when starting the stopwatch caused random error.”
“The results were good.”	“The repeats were close together, so the data is precise.”
“It's a bar chart.”	“A bar chart was used because material type is categoric.”
“Keep it a fair test.”	“Use the same trolley and the same start line each time.”

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FINISH

PROGRESS TRACKER

Colour one circle for each statement: **red** = not yet, **amber** = getting there, **green** = I can do this without help. Revisit your reds first.

I CAN…	R / A / G
01: Can this data be trusted?	
tell precision from accuracy	
calculate a mean and an uncertainty from repeats	
tell random from systematic error and say how to deal with each	
state the resolution of an instrument	
02: What are you actually changing?	
identify the independent, dependent and control variables	
say whether data is continuous or categoric	
choose a line graph or a bar chart and give the reason	

! **One habit worth building.** When something in your results looks wrong, the useful question is never “can I ignore that?” It is “why did that happen?”

THE WHOLE UNIT IN ONE SENTENCE

Measure it more than once, be honest about the doubt, show the pattern clearly, and let the evidence decide.

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CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in your teacher's answer edition.

LESSON 1: CAN THIS DATA BE TRUSTED?

DO NOW

1. Any sensible reason, e.g. your reaction time when starting and stopping the stopwatch is different each time, or the car is released slightly differently.
2. (4 + 5 + 6) ÷ 3 = 15 ÷ 3 = 5 cm
3. The one marked in millimetres.

COVER THE PAGE

1. Repeat readings are close to each other.
2. Close to the true value.
3. (largest − smallest) ÷ 2
4. Random error.
5. Systematic (a zero error).
6. 1 mm

LESSON 2: WHAT ARE YOU ACTUALLY CHANGING?

DO NOW

1. close to each other
2. random
3. (3.6 − 3.2) ÷ 2 = 0.2 s

COVER THE PAGE

1. The variable that is measured.
2. If two things change, you cannot tell which one caused the result.
3. Categoric.
4. The independent variable.

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REFERENCE

THE PHYSICIST'S TOOLKIT

Everything you need for the whole unit, on one page. Fold the corner: you will come back to it.

THE THREE VARIABLES

Independent	the one thing you change
Dependent	the thing you measure
Control	everything you keep the same

Ask: "How does *independent* affect *dependent*?"

TWO KINDS OF DATA

Continuous	any value on a scale: length, time, mass → **line graph**
Categoric	names or groups: material, colour → **bar chart**

Test: is there a value in between?

THE TWO CALCULATIONS

mean = total of the trials ÷ number of trials

uncertainty = (largest − smallest) ÷ 2

Write the result as mean ± uncertainty, e.g. 1.22 s ± 0.03 s.

PRECISE OR ACCURATE?

Precise	repeats close *to each other*
Accurate	close *to the true value*
Resolution	the smallest change an instrument can detect

You can be precise and still wrong.

THE TWO ERRORS

Random	scatters repeats both ways. **Repeat and take a mean.**
Systematic	shifts every reading the same way. **Find the cause; repeats do not help.**

CHOOSING A SCALE

Largest value ÷ number of big squares, then round **up** to 1, 2, 5, 10, 20 or 50 per big square. Equal steps. Use most of the grid.

ANOMALOUS RESULTS

Spot it (far from the trend) → **check** the recording, maths and method → **repeat** if you can. Mistake found: correct it. No cause: keep it, circle it, report it, and leave it out of the best-fit line. Never delete it silently.

CONCLUSION AND EVALUATION

Conclude: "As X increased, Y increased", with two values and units. Straight or curved?

Evaluate: a strength, a weakness, a named error, and an improvement that fixes it. Never just "human error".

GRAPH CHECKLIST

- Title: what against what
- Independent variable on the x-axis
- Both axes labelled, with units: "Ramp height / cm"
- Scale in equal steps that fills the grid
- Means plotted as small, accurate crosses
- One best-fit line: never dot-to-dot
- Bar chart: equal bars, equal gaps, axis from 0
- Anomalous results circled, or "none"

THE RULE THAT DECIDES YOUR GRAPH

Continuous independent variable → line graph. Categoric independent variable → bar chart.

It depends on what you *changed*, not on what you measured.

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